

Aakash Tirathdas

aakashtirathdas@gmail.com | +1 778-363-9094 | Kelowna, BC | [GitHub: aakash-tir](https://github.com/aakash-tir) | [LinkedIn](#)

Computer Science graduate from UBC Okanagan (Class of 2026, Data Science minor) with research experience in multi-agent LLM systems and applied ML, plus hands-on experience with Python, Java, and AWS. Seeking new-grad software engineering and ML roles. Open to relocation across Canada.

EDUCATION

University of British Columbia, Okanagan — BSc Computer Science, Minor Data Science Sep 2022 – Jun 2026

- Average: 83.2% (First Class Standing)

British School in Colombo — Cambridge A-Levels & IGCSEs Sep 2018 – Jun 2022

- A-Levels: A in Mathematics, A in Computer Science, A in Economics, B in Physics
- IGCSEs: 6 A* and 2 A grades across 8 subjects

EXPERIENCE

Research Assistant — University of British Columbia (Okanagan) Jan 2026 – Apr 2026

- Collaborated with MSc researchers on LLM faithfulness, reproducing results from recent papers and designing follow-up experiments.
- Conducted literature reviews to identify methodological gaps and propose extensions to the lab's ongoing work.
- Implemented experimental pipelines in Python and analyzed results to inform next research directions.

Research Assistant — Dr. Mohamed Afzal Norat Jul 2024 – Apr 2025

- Built multimodal and multi-agent LLM systems for enterprise task automation, targeting reduced computational cost and faster training/inference.
- Deployed and trained models on AWS EC2; managed cloud compute resources for experiments.

Computer Science Intern — CodeGen, Colombo, Sri Lanka Jul 2023 – Aug 2023

- Supported chatbot training pipeline in Python, including data preparation and labeling for classification tasks.
- Prototyped a proof-of-concept integrating the OpenAI API to ground chatbot responses in internal company data.

PROJECTS

Jersey Number Recognition — Deep Learning (COSC 419B) | Python, PyTorch | UBC Okanagan

- Modified a deep learning pipeline that processes image tracklets to identify player jersey numbers; improved baseline accuracy from 86.71% to 88.52% (+1.81 pp) by fine-tuning classification threshold parameters and adding Gaussian noise and motion blur augmentation.
- Investigated auxiliary heads on the Scene Text Recognition model as an additional improvement; analyzed the accuracy regression and reverted to the leaner configuration.

Blume — Project Analysis with Resume & Portfolio Generation (COSC 499 Capstone) | Python, React | UBC Okanagan

- Co-developed an MVP web app that ingests user code repositories and generates tailored resumes and portfolios from project history; shipped across three milestones with a team of six.
- Built the folder traversal and project-identification module from scratch, combining DFS/BFS into a single algorithm with a heuristic-based evaluator to detect project roots inside arbitrary uploaded archives — foundation for the team's analysis pipeline.
- Integrated the React upload frontend with the Python analysis backend end-to-end: designed a new database table tracking upload metadata, replaced the legacy fixed-path pipeline with dynamic path passthrough, and routed analysis results into the correct DB tables.
- Refactored the task analysis backend from single-task to multi-task support, unblocking multi-project analysis.

US Flight Delay Analysis (DATA 301) | Python, Tableau | UBC Okanagan

- Analyzed US domestic flight delays from 2019–2021 using statistical modeling and visualizations to identify carrier- and route-level patterns.
- Built the analysis in Jupyter Notebooks with Pandas, Seaborn, and Matplotlib; presented findings via Tableau dashboards.

Game of Amazons AI Bot (COSC 322) — Two-player abstract strategy game | Java | UBC Okanagan

- Implemented Minimax with alpha-beta pruning and designed custom heuristics for board evaluation.
- Integrated with the course-provided Java library and game server for matches against other student bots.

SKILLS

Languages: Python, Java, JavaScript, SQL, R, HTML/CSS

Frameworks & Libraries: PyTorch, React, Pandas, NumPy, Matplotlib, Seaborn

AI / ML: Deep Learning, Statistical Analysis, RAG systems, Multi-agent LLM systems, Prompt Engineering

Tools & Infrastructure: AWS (EC2), Docker, Git, Jupyter, Tableau, REST APIs